

CSharPN

A Code-First Framework for Coloured Petri Net Modelling
with C# as Inscription Language

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Underneath it, the older gap between **academia and industry**.

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CSharPN — a framework for working with Coloured Petri Nets in C#, with a VS Code extension for visualising and executing models.

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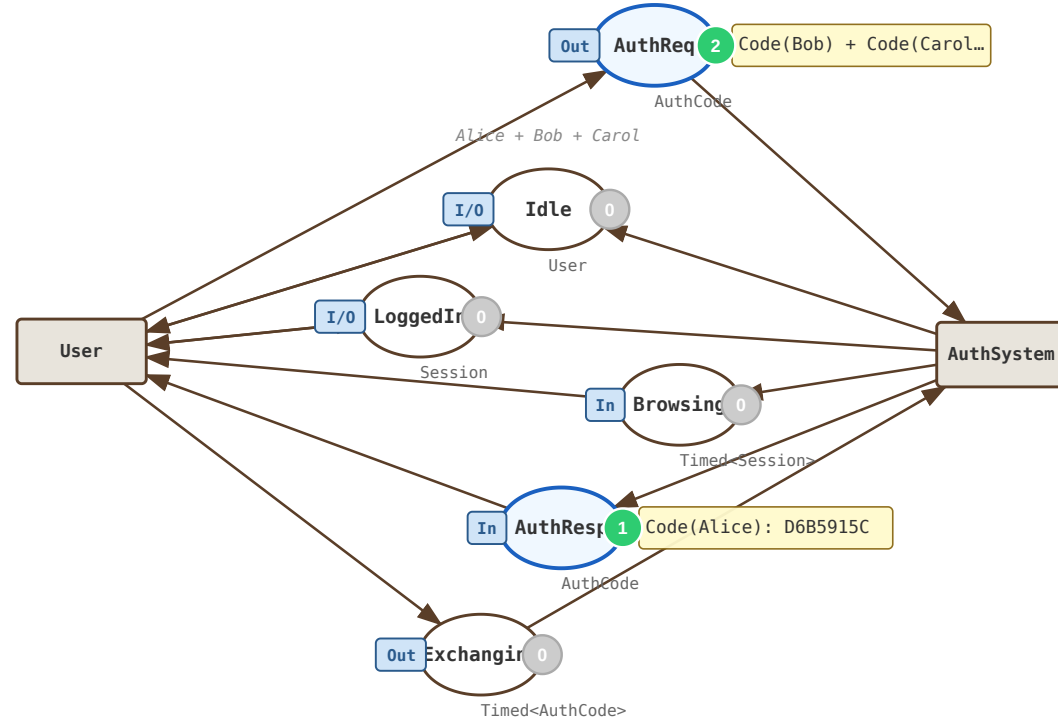
Simulation-based development — with concurrency, resource sharing, timing, and hierarchy modelled *explicitly*.

Running example:

OAuth2 authorisation code flow

Exercises all four features:

Top page: shared places link the two substitution transitions.



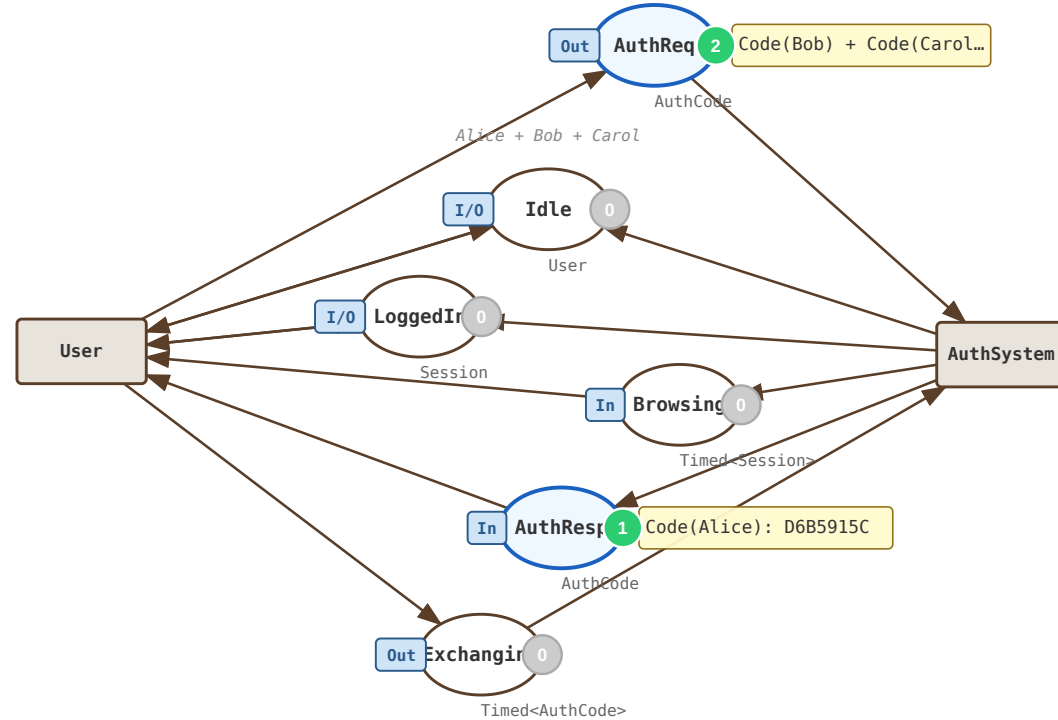
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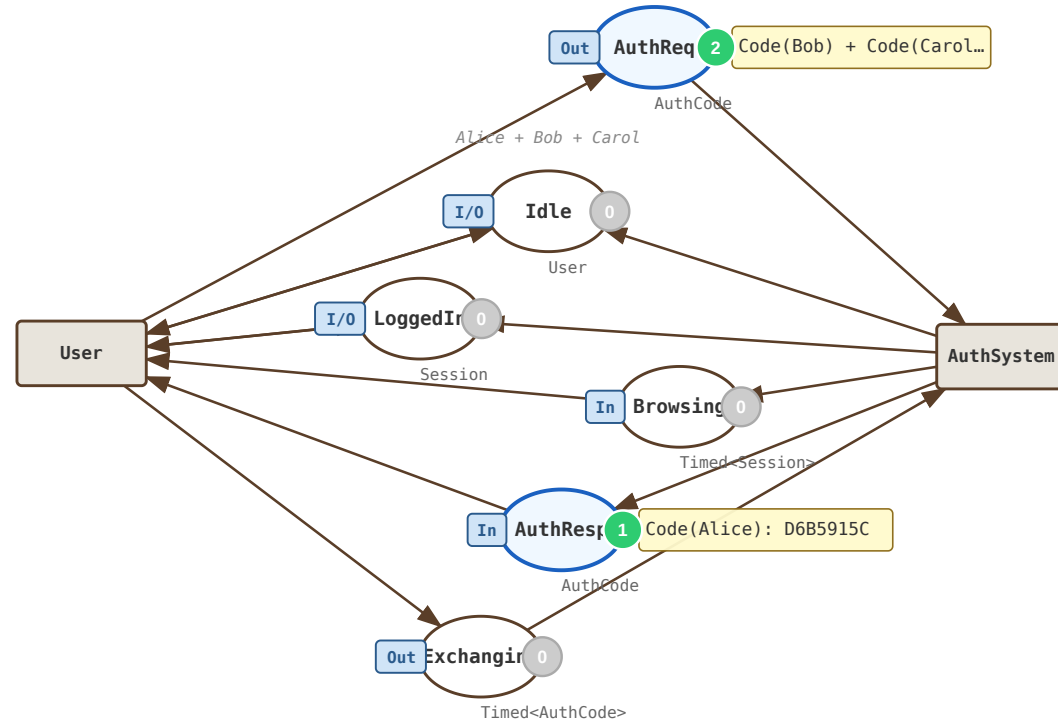
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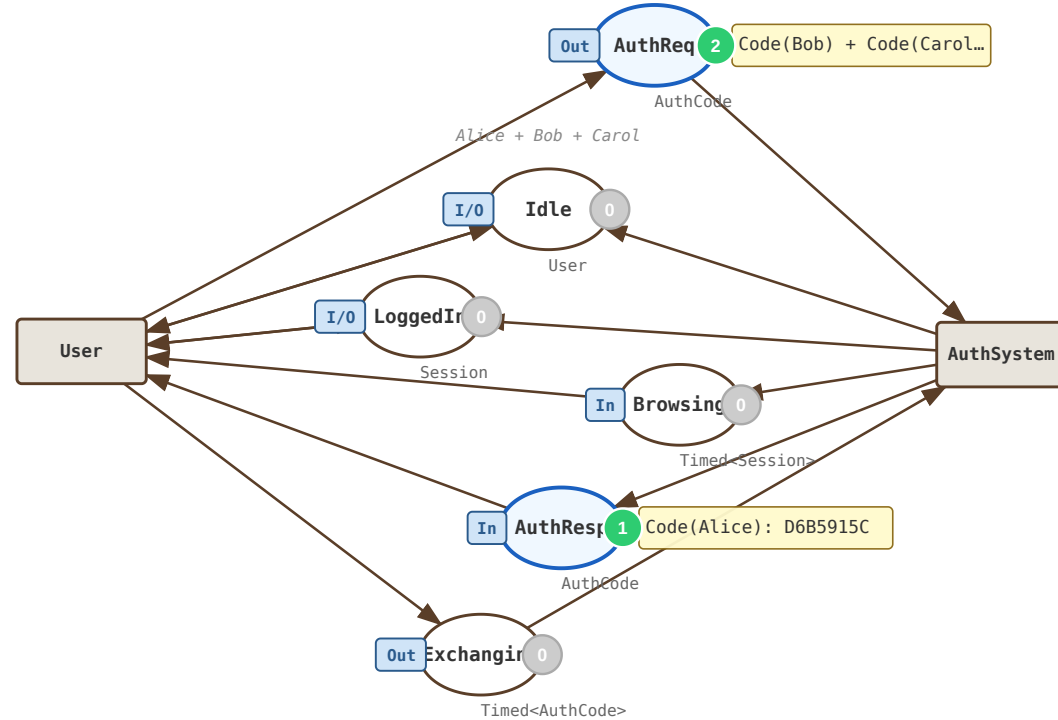
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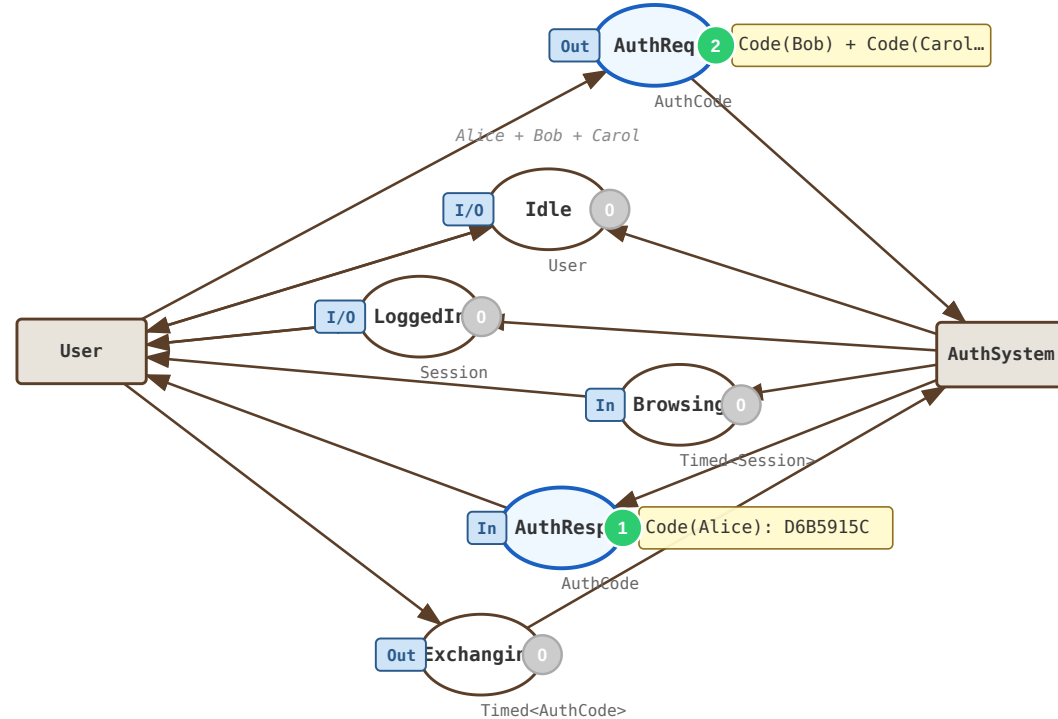
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- Timed auth, exchange, expiry
- Two sub-pages: User & AuthSystem

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Colour sets are just C# types

Any `notnull, IEquatable<T>` type works — primitives, enums, records, classes.

```
public record User(int Id, string Name) {  
    public string CsrfsState => $"s{Id}";  
    public int    BrowseTime => Id * 5;  
}  
  
public record AuthCode(User User, string Code, string State) {  
    public bool ValidCsrfs(User u) => State == u.CsrfsState;  
    public static AuthCode Create(User u, int seq) {  
        var hash = SHA256.HashData(Encoding.UTF8.GetBytes($"{u.Id}:{seq}"));  
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- **Records** — immutable, with value equality
- Methods carry **domain logic** (`CsrfsState` , `BrowseTime`)

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Places & multisets

```
// inside the model constructor
var authReq = AddPlace<AuthCode>("AuthReq");           // empty initial marking

var idle    = AddPlace("Idle", Multiset.Of(
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Transitions: a fluent builder

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var u = new Var<User>("u");

AddTransition("Login")
    .Input(idle, u).Output(waiting, u)
    .Output(authReq, () => AuthCode.Create(u.Val), "AuthReq")
    .Build();

AddTransition("ReceiveCode")
    .Input(authResp, c).Input(waiting, u)
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Timed CPNs

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AddTransition("GrantToken")
    .TimedInput(exchg, c).Input(nextTokId, tid)
    .Output(nextTokId, () => tid.Val + 1)
    .Output(loggedIn, () => new Session(c.Val.User, tid.Val))
    .TimedOutput(expiry, () => new Session(c.Val.User, tid.Val), Config.SessionTTL)
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    .Build();

AddTransition("SessionExpire")
    .TimedInput(expiry, se).Input(loggedIn, se2)
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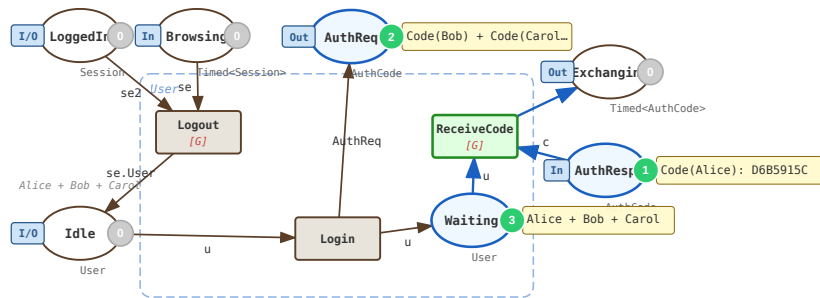
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Hierarchical CPNs

```
class UserPage : CpnPage {
    public UserPage(Place<User> idle,
        Place<AuthCode> authReq, ...)
        : base("User") {
        InOut(idle);
        Out(authReq); In(authResp);
        Out(exchg);
        InOut(loggedIn); In(browsing);
        var waiting = AddPlace<User>("Waiting");
        // Login, ReceiveCode, Logout
    }
}
```

```
AddSubPage(new UserPage(...), "User");
AddSubPage(new AuthSystemPage(...), "AuthSystem");
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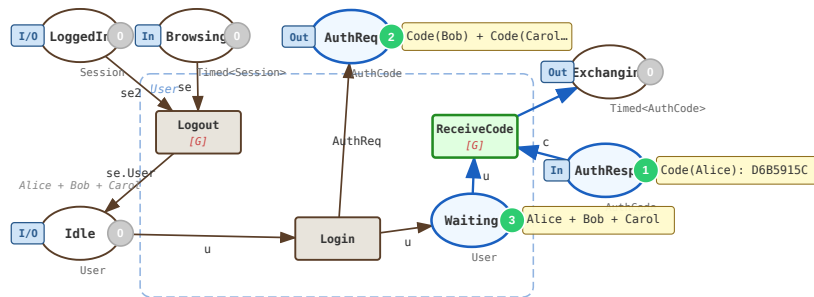


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- A sub-page is a `CpnPage` **subclass**

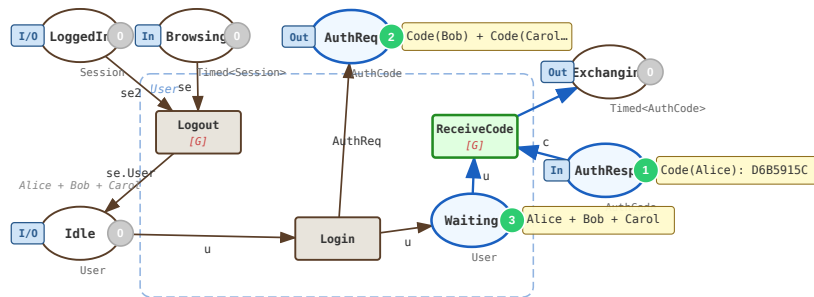
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- A sub-page is a `CpnPage` **subclass**
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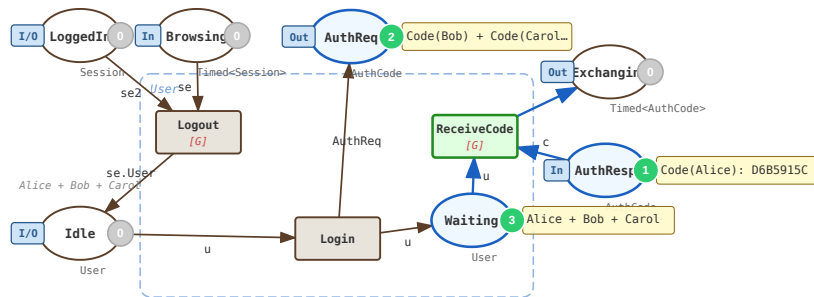


Hierarchical CPNs

```
class UserPage : CpnPage {  
    public UserPage(Place<User> idle,  
        Place<AuthCode> authReq, ...)  
        : base("User") {  
        InOut(idle);  
        Out(authReq); In(authResp);  
        Out(exchg);  
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AddSubPage(new UserPage(...), "User");  
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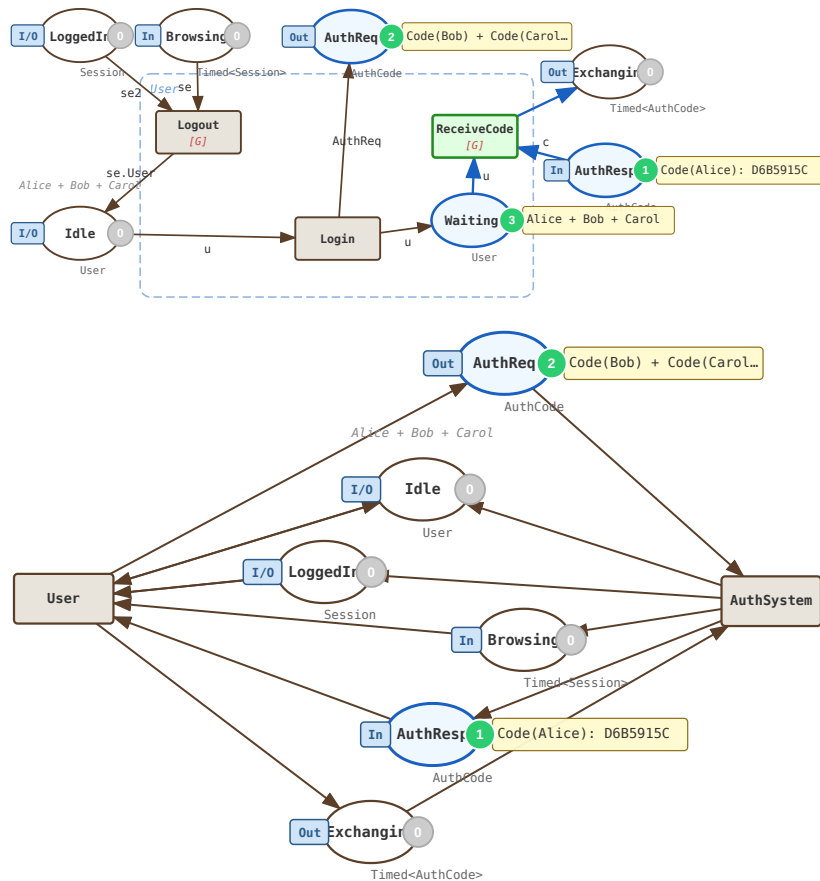


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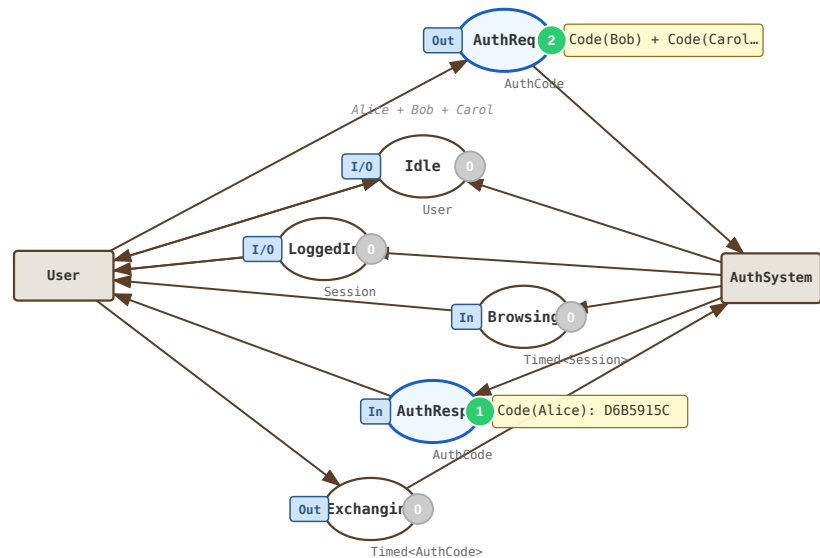
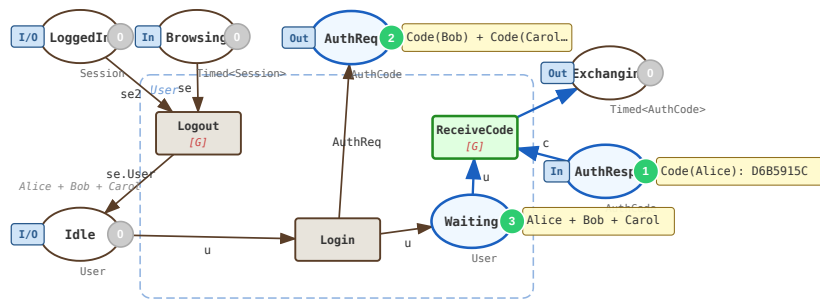


Hierarchical CPNs

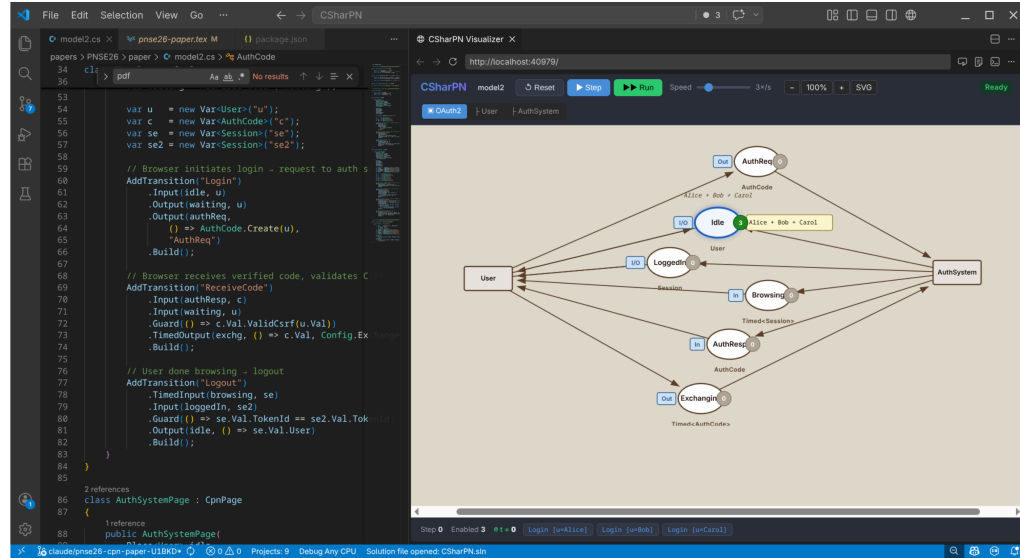
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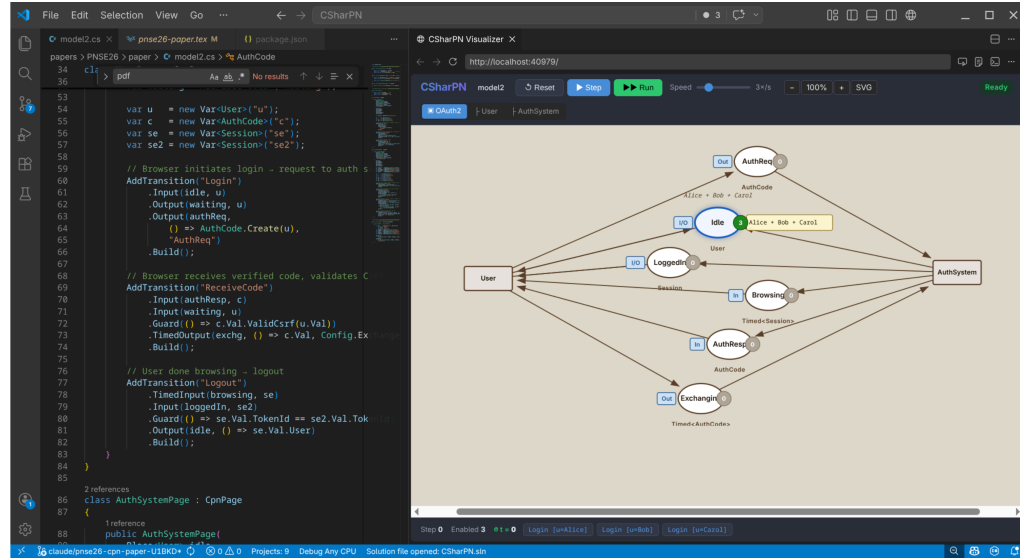


Visualiser & VS Code integration



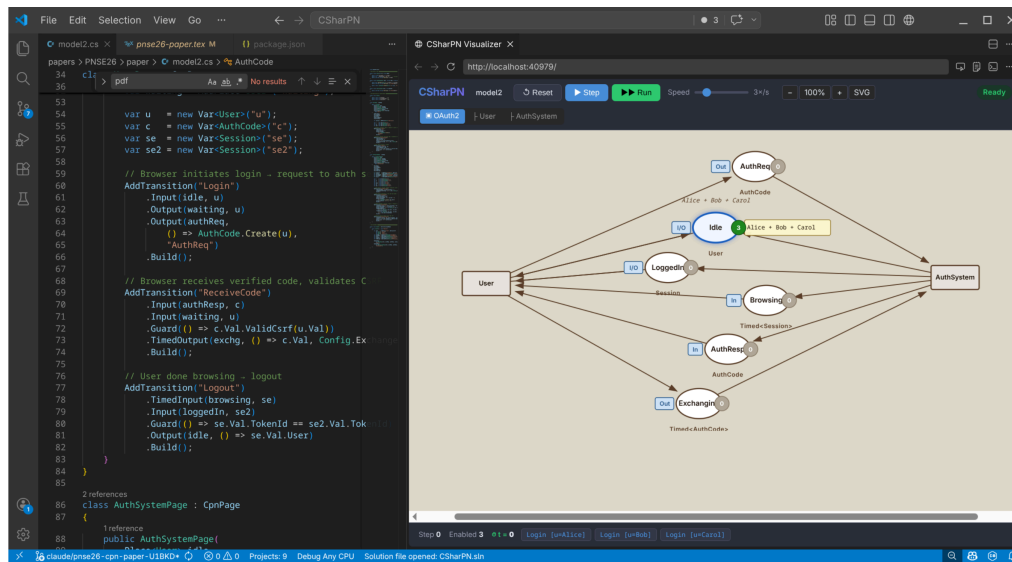
Visualiser & VS Code integration

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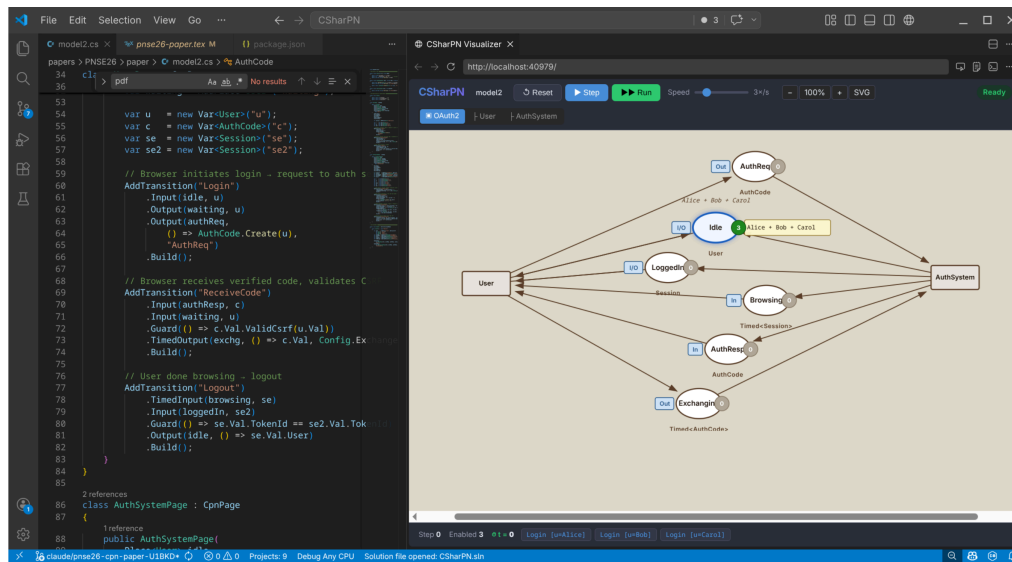
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- Page tree — drill into sub-pages



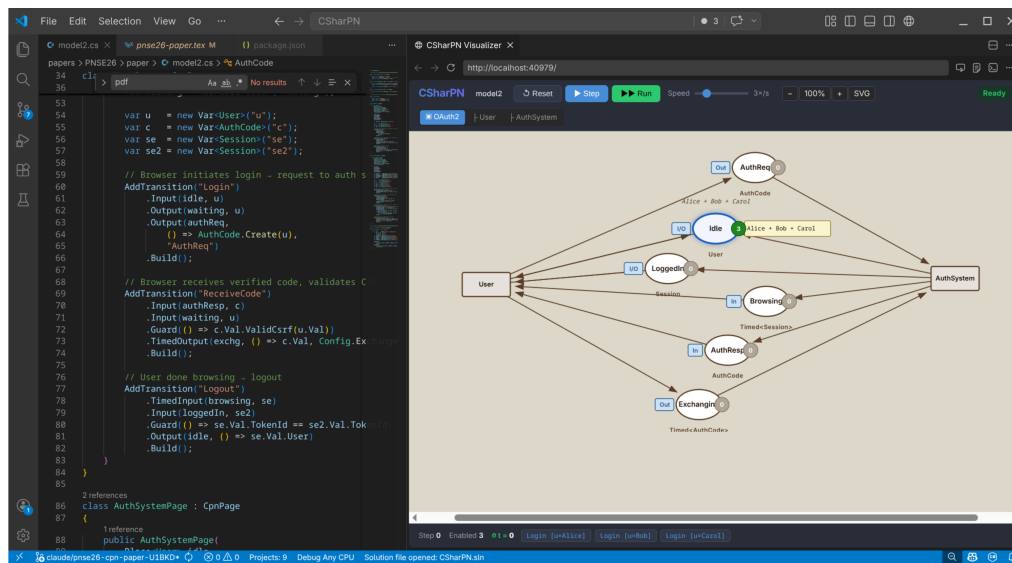
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- Blazor app in VS Code's Simple Browser panel
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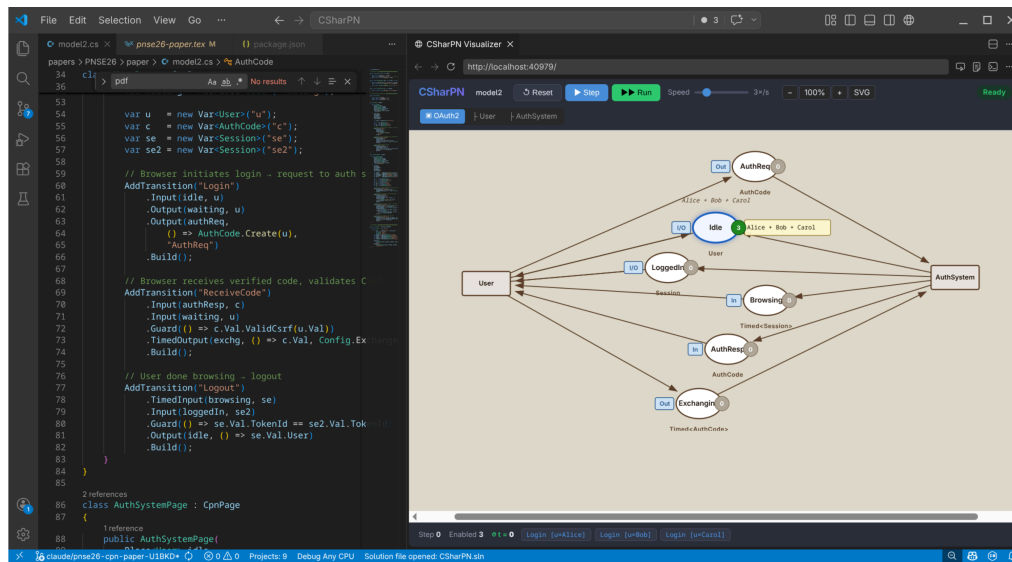
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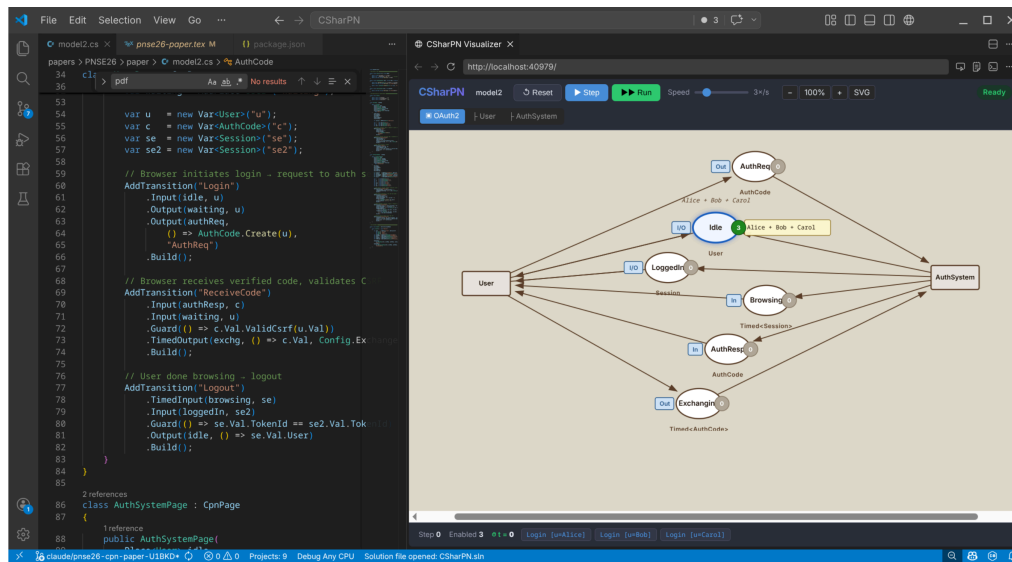
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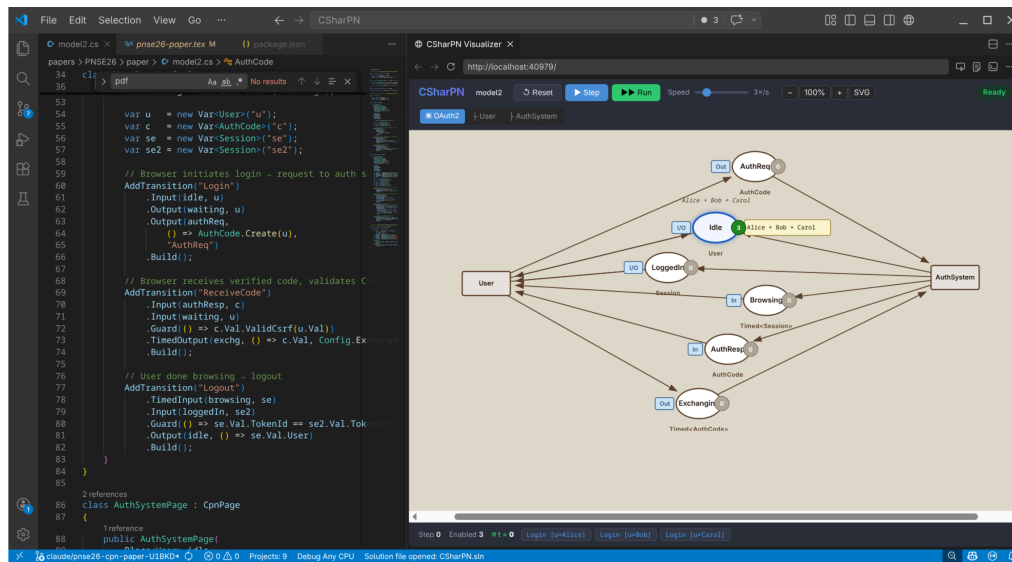
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- Layout persisted in `model.cs.layout`
- **Hot reload**: save .cs → recompile → reload, marking preserved



Future work

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- State-space analysis on `CpnState` snapshots

Thank you

Questions?



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github.com/simontjell/CSharpN.public

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